Boosting Sales at A.R Furniture through Strategic Inventory Optimization and Customer-Centric Design

Final report for the BDM capstone Project

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Table of Contents

1. **Executive Summary** **3**
2. **Detailed Explanation of Analysis Process/Method** **3-13**
   - 2.1 Introduction
   - 2.2 Data Cleaning and Pre-processing
   - 2.4 Demand Stability Metric (DSI)
   - 2.5 Improved ABC Classification – Full Methodology and Rationale
   - 2.6 Demand Trend Classification
   - 2.7 Demand Trend Tagging
   - 2.8 Final Decision Table (ABC Class × Demand Trend → Action)
3. **Results and Findings**
   - 3.1 Improved ABC Classification — Summary
   - 3.2 Top SKUs by ABC Score
   - 3.3 Demand Trend Tagging – Summary
   - 3.4 Forecasts (3-Month Moving Average) – Selected Examples
   - 3.5 Visualization
   - 3.6 Final Consolidated Action Table (SKU-Level)
   - 3.7 Summary of Key Findings
4. **Interpretation of Results and Recommendation** **20-**
   - 4.1 Strategic Implications of ABC Classification
   - 4.2 Operational Risk in Declining SKUs
   - 4.3 Demand Trend as a Planning Signal
   - 4.4 Reliability of 3-Month Moving Averages
   - 4.5 Integrated Action Planning

1. Executive Summary:

A.R. Furniture is a small business in Tiwari Tank Road, Ranchi, Jharkhand, 834001 that supplies raw furniture frames, components, and a limited selection of finished goods to both B2B and walk-in customers. The business runs on manual records and informal decision-making, which creates challenges like irregular purchasing and unclear product-level performance. This project aimed to turn scattered sales and purchase data into insights to support smarter inventory and procurement decisions.

Four months of raw and retail sales data were analyzed in Excel using ABC classification, Demand Stability Index (DSI), trend labeling, and profitability segmentation. These tools helped uncover which SKUs drive profit, how consistent demand is, and where inefficiencies lie.

The analysis revealed that 11 Class A SKUs generate over 87% of raw sales profit, highlighting the need to focus on a smaller, high-impact product set. Trend analysis using 3-month moving averages (SMA_1 and SMA_2) and coefficient of variation (CV) showed that many A-class items like SSF (A), SSF (B), and Teak Sawn Timber have stable or rising demand, while others like SSF (C) and SSF (D) show unpredictable or tapering trends. Items like BSF (B) and BSF (A) had high demand variability and multiple months with zero sales.

A.R. Furniture should prioritize Class A items in purchasing, with regular reviews and sufficient stock for high performers. SKUs with unstable or declining trends, even if profitable, should be closely monitored or limited. Class B items should be stocked cautiously, especially slow movers like BSF (A). Class C items with low profit and unstable demand like SSF (E) and BSF (B) should be phased out or discontinued. These steps will reduce overstock, free up working capital, and improve overall profitability.

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2. Detailed Explanation of Analysis Process and Methodology

2.1 Analytical Framework and Rationale

The analytical process was designed to move beyond profit-only classification toward a risk-adjusted inventory prioritization model.

Traditional inventory classification focuses primarily on financial contribution. However, inventory planning decisions also depend on demand stability, sales regularity, and short-term direction of movement. Therefore, this study adopted a structured, multi-stage framework integrating financial, statistical, and operational indicators.

The methodology consisted of:

- Data aggregation and validation
- Demand variability measurement
- Traditional ABC classification
- Multi-Criteria ABC modelling
- Demand trend classification and policy mapping

This layered approach ensures that inventory decisions reflect both profitability and operational risk.

2.2 Data Aggregation and Preparation

Transaction-level data (April–July 2025) were aggregated at SKU × Month level to create time-series demand profiles.

Monthly aggregation was necessary because:

- Variability metrics require periodic demand observations.
- Moving averages require consistent time intervals.
- Stability analysis requires comparable monthly demand values.

The preprocessing stage ensured numeric consistency and removed structural data inconsistencies to prevent distortion in standard deviation and normalization calculations.

This step improves analytical reliability but does not influence classification logic.

2.3 Demand Variability Measurement (Coefficient of Variation)

2.3.1 Justification for Using CV

Inventory control decisions depend heavily on demand predictability. Items with unstable demand require higher safety stock and tighter monitoring.

Standard deviation alone is insufficient because it measures absolute variability and is sensitive to demand scale. Therefore, the Coefficient of Variation (CV) was used as a scale-independent measure of relative variability.

$$CV = \frac{\sigma}{\mu}$$

Where:

- σ = Standard deviation of monthly demand
- μ = Average monthly demand

CV allows comparison across high-volume and low-volume SKUs, making it suitable for multi-item inventory analysis.

2.3.2 Operational Interpretation

- CV < 0.5 → Stable demand
- 0.5–1.0 → Moderate variability
- CV > 1 → Highly volatile / intermittent demand

Items with CV greater than 1 exhibit variability exceeding average demand, indicating elevated forecasting uncertainty.

Using CV aligns the model with established inventory management literature and risk-based stocking principles.

2.4 Traditional ABC Classification

2.4.1 Rationale

Traditional ABC classification is grounded in Pareto's principle, which states that a small proportion of items typically contribute the majority of value.

This method prioritizes SKUs based on total profit contribution.

Classification thresholds (70–20–10) are commonly used in operations management and reflect practical segmentation standards.

2.4.2 Method

Traditional ABC classification was performed based on Total Profit Contribution.

Steps:

- SKUs ranked in descending order of Total Profit.
- Profit Contribution calculated:

$$\text{Profit Contribution}_i = \frac{\text{Total Profit}_i}{\sum \text{Total Profit}}$$

- Cumulative percentage computed.
- Classification applied:
    - Class A → Top ~70% cumulative contribution
    - Class B → Next ~20%
    - Class C → Remaining

2.4.3 Limitation

While effective for financial prioritization, this method does not differentiate between stable and volatile items. As a result, high-profit but erratic SKUs may be over-prioritized.

2.5 Multi-Criteria ABC Classification

2.5.1 Rationale for Model Extension

To address limitations of single-criterion classification, a composite ABC Score was developed integrating:

- Financial impact (Profit)
- Demand regularity (Sales Frequency)
- Demand predictability (Stability derived from CV)

This ensures classification reflects both economic contribution and operational reliability.

2.5.2 Justification for Feature Selection

- Profit reflects revenue importance.
- Sales frequency reflects transaction regularity and demand continuity.
- Stability reflects predictability and safety stock sensitivity.

Together, these metrics provide a balanced representation of SKU importance.

Sales Frequency was selected instead of total quantity because quantity alone may represent bulk transactions rather than consistent demand. Frequency captures how regularly a SKU is sold, which is operationally more relevant for replenishment planning.

Demand Stability was derived from the Coefficient of Variation (CV), which measures variability relative to average demand. CV is preferred over standard deviation because it allows comparison across SKUs with different demand scales.

2.5.3 Normalization

Min–Max normalization was applied:

$$X_{norm} = \frac{X_i - Min(X)}{Max(X) - Min(X)}$$

This standardizes all features between 0 and 1.

2.5.4 Normalization Justification

Because metrics operate on different scales (profit in currency, frequency in counts, CV as ratio), Min–Max normalization was applied to ensure comparability and prevent scale dominance.

Demand stability was obtained by inverting normalized CV:

$$Stability = 1 - CV_{normalized}$$

This transformation ensures consistent directional interpretation: higher score indicates better performance.

2.5.6 Composite ABC Score

$$ABC\ Score = 0.4 \times Normalized\ Profit + 0.3 \times Normalized\ Sales\ Frequency + 0.3 \times Normalized\ Stability$$

Profit was weighted higher (40%) due to direct financial impact, while frequency and stability were each weighted 30% to reflect operational reliability.

2.5.7 Weighting Rationale

The weighting scheme (0.4 / 0.3 / 0.3) reflects:

- Slightly higher emphasis on financial contribution
- Equal importance to operational reliability and regularity

This balanced weighting avoids over-dominance of any single factor.

2.5.8 Classification Threshold Justification

A 70% cumulative threshold was retained to maintain comparability with traditional ABC standards while reflecting composite importance.

This ensures operational alignment without distorting industry classification norms.

2.6 Demand Trend Analysis

2.6.1 Objective

Trend tagging was performed to convert static ABC classification into actionable inventory rules.

2.6.2 Method

Two moving averages were computed:

$$SMA_1 = \text{Average (Apr, May, Jun)}$$
$$SMA_2 = \text{Average (May, Jun, Jul)}$$

Trend determined using thresholds:

- $SMA_2 > 1.10 \times SMA_1 \rightarrow$ Rising
- $SMA_2 < 0.90 \times SMA_1 \rightarrow$ Declining
- Within ±10% → Stable

A ±10% threshold was selected to filter minor noise while capturing material demand shifts.

2.6.3 Volatility Override

If:

- CV > 1

  or
- ≥ 2 zero-sale months

Then SKU classified as **Unpredictable**, overriding trend classification.

This rule is consistent with intermittent demand literature (Syntetos & Boylan, 2005).

2.7 Final decision table (ABC class × Demand trend → Action) Create a compact decision table (paste into report). Example:

Table 2.7.1: Inventory Action Matrix (ABC × Trend)

ABC CLASS	DEMAND TREND	ACTION
A	Rising	Increase reorder frequency; raise safety stock; ensure supplier lead time is short.
A	Stable	Maintain optimal reorder point; periodic review; prioritize supplier reliability.
A	Declining	Reduce order size gradually; investigate demand cause; do not overstock.
B	Rising	Monitor daily/weekly; increase review frequency; consider promotional push.
B	Stable	Controlled replenishment; regular review.
C	Rising	Evaluate margin; if profitable, consider moving to B monitoring.
C	Stable/Declining	Minimize stock; order only on demand or vendor dropship; phase out if non-profitable.

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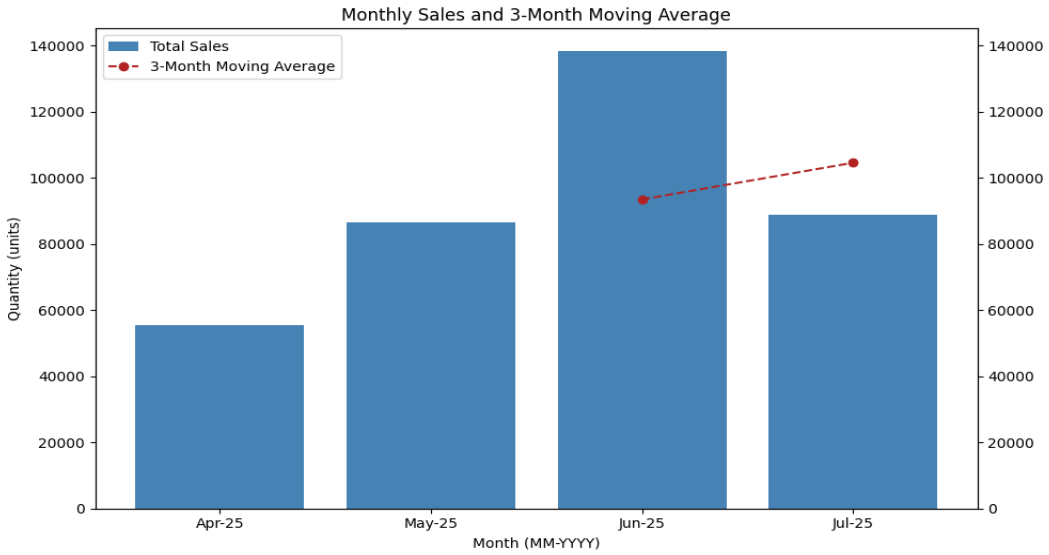
3. Results and Findings

3.1 Overall Sales Performance (April–July 2025)

Figure 3.1 Monthly Sales and 3-Month Moving Average



The monthly sales trend indicates progressive growth from April to June 2025, followed by marginal stabilization in July. The peak observed in June is primarily driven by significant volume movement in Teak Sawn Timber and select frame components, rather than uniform growth across all SKUs.

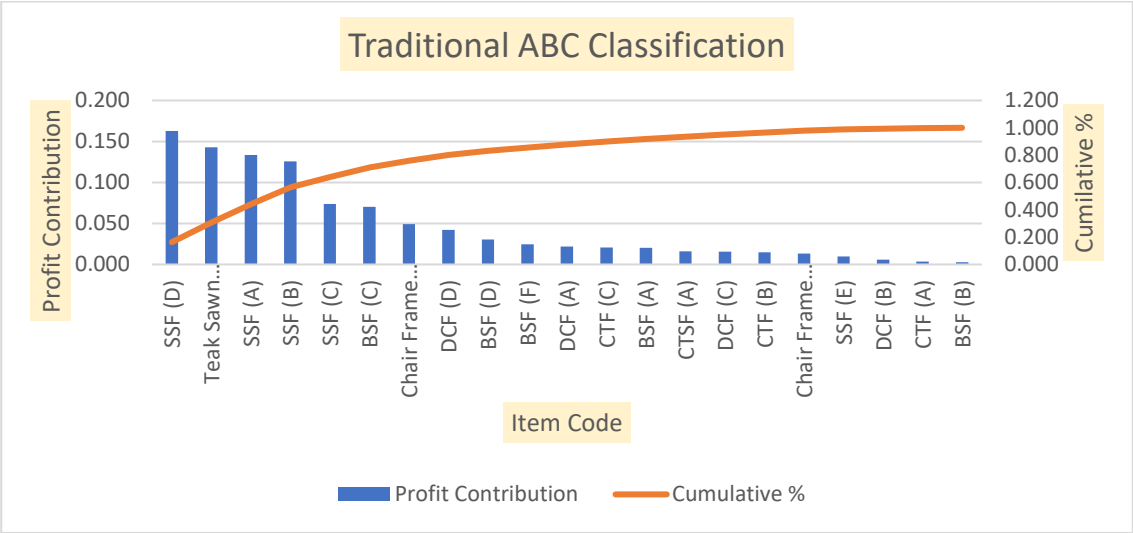
The three-month moving average confirms mid-period acceleration in demand, suggesting temporary demand expansion rather than sustained structural growth.

However, SKU-level inspection reveals that several items exhibit zero-sale months, indicating intermittent purchasing behaviour rather than continuous consumption patterns.

This uneven demand distribution justifies the need for structured classification and risk-adjusted inventory planning.

3.2 Traditional ABC Classification (Profit-Based)

Figure 3.2 Traditional ABC Classification Graph



The traditional ABC classification was conducted by ranking SKUs in descending order of total profit and computing cumulative profit contribution.

From the provided table:

- The top SKU, **SSF (D)**, contributes **16.3%** of total profit.
- The top two SKUs together contribute **30.6%**.
- The top four SKUs together contribute **56.5%**.
- The cumulative contribution reaches **80.0% at DCF (D)**.
- The first **7 SKUs** account for **75.8%** of total profit.
- Class A ends at the 7th SKU (Chair Frame A, cumulative 75.8%).
- Class B extends from cumulative 80.0% to 89.8%.
- Class C contains the remaining low-contribution SKUs.

Final classification (as per your table):

- **Class A:** 7 SKUs (up to 75.8% cumulative)
- **Class B:** 5 SKUs (75.8%–89.8%)
- **Class C:** 9 SKUs (remaining 10.2%)

Key Findings:

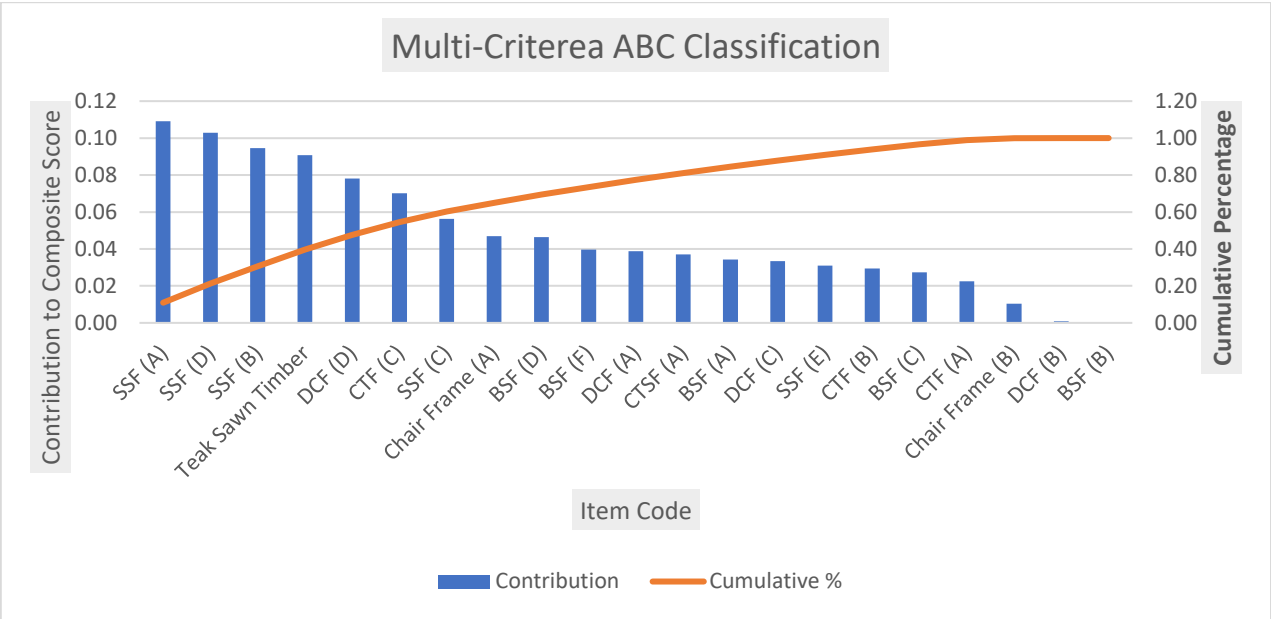
- Profit concentration is strong but not extreme — the top 7 items generate approximately three-fourths of total profit.
- Individual contribution drops sharply after SSF (B), indicating diminishing marginal impact.
- Class C contains many SKUs with very low individual profit contribution (<2%).

Interpretation:

Traditional ABC clearly identifies revenue-dominant SKUs but does not incorporate demand stability, sales frequency, or risk exposure. The classification is purely financial and reflects revenue concentration structure.

3.3 Multi-Criteria ABC Classification (Composite Model)

Figure 3.3 Multi-Criteria ABC Classification Graph



The multi-criteria ABC classification ranks SKUs based on a composite score derived from:

- Normalised Profit
- Normalised Sales Frequency
- Normalised Stability (1 − CV normalised)

The SKUs are sorted in descending order of **ABC Score contribution**, and cumulative percentage is calculated accordingly.

3.3.1 Findings:

From the cumulative % column:

- SSF (A) alone contributes **11%**
- Top 3 SKUs contribute **31%**
- Top 5 SKUs contribute **48%**
- Top 7 SKUs contribute **60%**

- Cumulative reaches **70% at BSF (D)**
- Cumulative reaches **80% at CTSF (A)**

Based on your classification column:

- **Class A:** First 9 SKUs (up to 70% cumulative)
- **Class B:** Next 5 SKUs (70%–88%)
- **Class C:** Remaining SKUs (above 88%)

Total:

- Class A = 9 SKUs
- Class B = 5 SKUs
- Class C = 7 SKUs

(This matches your table exactly.)

3.3.2 Key Observations

1. The cumulative curve rises more gradually compared to traditional ABC.
2. Importance is more evenly distributed across top SKUs.
3. Some high-profit items move lower due to high CV (low stability score).
4. Stable items such as CTF (C) improve ranking despite moderate profit.
5. Items with CV = 2 receive zero stability score and fall to Class C.

3.3.3 Interpretation

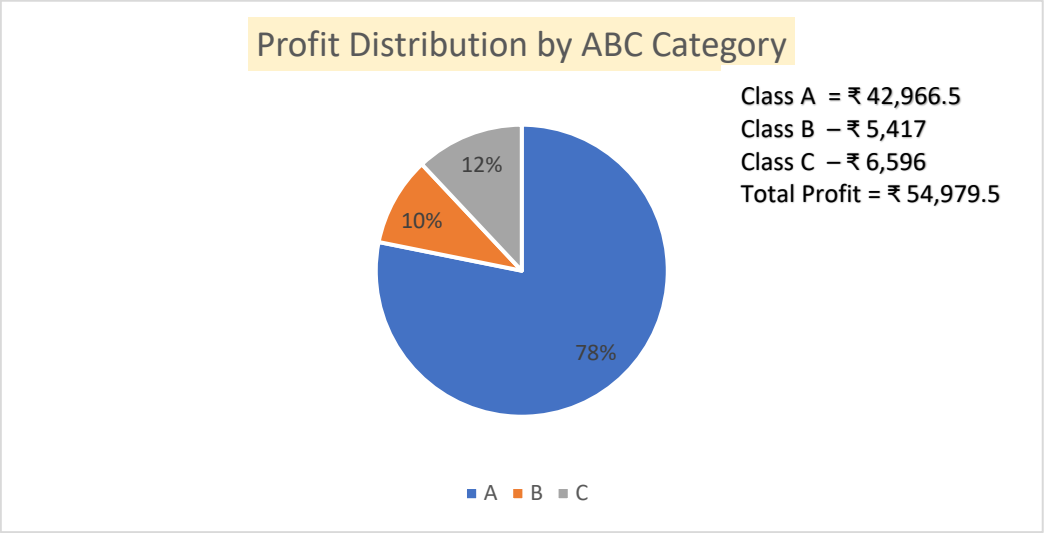
Unlike traditional ABC (which concentrates purely on financial weight), the multi-criteria model:

- Penalises demand volatility
- Rewards sales frequency
- Rewards stability
- Produces a risk-adjusted prioritisation

This results in a broader and more operationally balanced Class A group

3.4 Profit Distribution by Multi-Criteria ABC Category

Figure 3.2 Profit Distribution Graph:



The profit contribution by ABC class is presented in Figure 3.2 (Pie Chart).

Class A items contribute approximately **78.15% of total profit**, despite representing only 9 out of 21 SKUs. Class B contributes about 9.85%, while Class C contributes 12%.

This confirms strong Pareto concentration within the inventory portfolio, where a limited number of SKUs generate the majority of financial value. The classification therefore justifies prioritised monitoring and tighter inventory control for Class A items.

3.4 Demand Trend Classification and Findings

3.4.1 High Demand Irregularity

The majority of SKUs were classified as Unpredictable, driven by:

- CV values greater than 1
- Multiple zero-sale months

This indicates intermittent and unstable demand across a significant portion of the inventory.

3.4.2 Limited Truly Stable Items

Only **CTF(C)** qualified as Stable (CV = 0.61 with consistent movement). This suggests that steady, low-variance demand is rare in the current SKU mix.

3.4.3 Identified Rising SKUs

- DCF (D)
- SSF (A)
- SSF (B)

This show sustained upward movement with acceptable volatility levels.

3.4.4 Identified Declining SKU

- SSF (D)

SMA_2 was meaningfully lower than SMA_1, indicating contraction.

3.4.5 Borderline Observations

Some SKUs appear directionally increasing but remain classified as Unpredictable due to high CV or multiple zero months.

This reflects short data horizon sensitivity and reinforces the need to prioritise statistical consistency over visual interpretation.

3.4.6 Overall Insight

Demand behaviour is structurally volatile, even among several Class A SKUs.

This implies that inventory control decisions cannot rely on profitability alone and must incorporate demand stability.

The revised classification provides a disciplined and reproducible approach to distinguishing:

- Directional growth
- True stability
- Intermittent unpredictability

3.5 Safety Stock and Reorder Level Insights

Safety stock was computed using service-level-based Z-values derived from the standard normal distribution. The Z-score represents the number of standard deviations required to achieve a specified probability of avoiding stockout during lead time. In alignment with inventory criticality under ABC classification, differentiated service levels were applied: Class A items were assigned Z = 1.88 (≈97%

service level), Class B items Z = 1.65 (≈95% service level), and Class C items Z = 1.28 (≈90% service level). This differentiation ensures that financially critical items receive higher protection against demand variability, while lower-impact items operate under leaner inventory policies. The approach reflects a risk-adjusted inventory control strategy, balancing service performance and working capital efficiency:

- Class A → Z = 1.88
- Class B → Z = 1.65
- Class C → Z = 1.28

Reorder Level = Average Demand + Safety Stock

3.5.1 Key Observations

1. Protective Buffer Dominance in Volatile SKUs

For several high-CV items, safety stock exceeds or closely matches average demand:

- BSF (C):

  Average = 1, Safety Stock = 3

- DCF (C):

  Average = 4, Safety Stock = 10

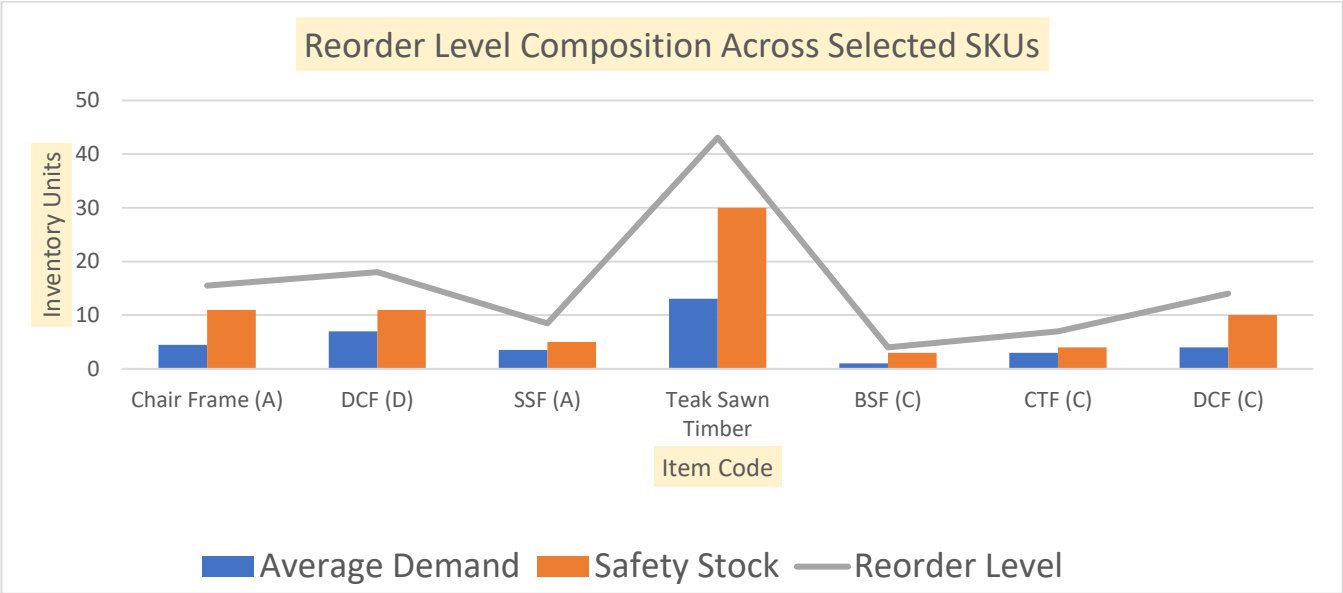
- Teak Sawn Timber:

  Average = 13.08, Safety Stock = 30

In such cases, the majority of inventory is buffer stock rather than cycle stock.

2. Working Capital Sensitivity in Class A Items

Figure 3.5 Reorder Level Composition Across Selected SKUs



Class A SKUs carry significantly larger buffers due to higher service levels:

- DCF (D): Safety Stock = 11
- Chair Frame (A): Safety Stock = 11
- Teak Sawn Timber: Safety Stock = 30

This creates strong capital lock-in in volatile but critical items.

3. Efficient Inventory in Stable SKUs

Low-CV items show relatively controlled buffers:

- CTF (C): Average = 3, Safety Stock = 4
- SSF (A): Average = 3.5, Safety Stock = 5

Here, inventory is more demand-aligned and less risk-driven.

3.5.2 Strategic Interpretation

The analysis reveals:

- High variability drives disproportionately high safety stock
- Class A volatility significantly influences capital allocation
- Intermittent SKUs require cautious replenishment strategies
- Stable SKUs allow leaner inventory positioning

Overall, inventory risk is structurally embedded within demand variability rather than solely financial ranking.

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4. Interpretation of Results and Recommendations

4.1 Integrated Inventory Risk Profile

The combined analysis reveals that inventory risk at A.R. Furniture is not driven solely by financial concentration, but by the interaction between profitability, demand variability, and trend behaviour.

Traditional ABC analysis confirms revenue concentration within a limited set of SKUs. However, the multi-criteria and variability analysis shows that several financially critical SKUs exhibit moderate to high demand uncertainty. This creates structural exposure where high-value items also require substantial safety buffers.

The demand trend classification further indicates that a significant portion of the portfolio is classified as "Unpredictable," reflecting intermittent or irregular monthly demand patterns. Only a limited number of SKUs demonstrate stable behaviour with low variability.

Thus, inventory exposure is multidimensional:

- Financial concentration risk
- Variability-driven operational risk
- Capital lock-in through safety stock

This confirms that a uniform replenishment policy would be suboptimal.

4.2 Strategic Segmentation of SKUs

To translate analytical findings into operational guidance, SKUs were segmented based on ABC class, variability (CV), trend, and safety stock ratio.

4.2.1 Critical and Volatile SKUs

Examples: Teak Sawn Timber, Chair Frame (A)

Characteristics:

- Class A financial importance
- CV > 1
- High safety stock ratios (~0.70)
- Unpredictable trend

Interpretation:

These items combine revenue criticality with demand uncertainty, resulting in significant capital allocation toward buffer stock.

Recommendation:

- Continuous review system
- Monthly forecast updates
- Supplier lead-time reduction initiatives
- Quarterly safety stock recalibration

4.2.2 Critical but Stable SKUs

Examples: CTF (C), SSF (A)

Characteristics:

- Class A
- CV < 0.8
- Lower safety stock ratio (~0.57–0.59)
- Stable or Rising trend

Interpretation:

These items are financially important yet operationally predictable.

Recommendation:

Maintain high service level

Gradually optimise safety stock downward

Lean replenishment strategy

These SKUs offer opportunity for working capital efficiency without compromising service.

4.2.3 Moderate Value but High Variability SKUs

Examples: DCF (C), BSF (C)

Characteristics:

- Class B or C
- CV ≥ 1.4 (some at 2.0)
- High safety stock ratio (~0.75)
- Unpredictable trend

Interpretation:

Although financially less significant, these SKUs consume disproportionate safety buffers relative to their value contribution.

Recommendation:

- Periodic review system
- Lower service level tolerance
- Avoid excess stocking
- Consider minimum order or on-demand replenishment

This prevents capital being locked in low-impact volatile items.

4.2.4 Declining SKUs

Example: SSF (D)

Characteristics:

- Class A
- Low CV
- Declining trend

Interpretation:

Despite financial importance, declining momentum increases risk of overstock.

Recommendation:

- Gradual reduction in reorder level
- Conservative replenishment
- Monitor for structural demand shift

4.3 Working Capital Implications

The safety stock ratio analysis indicates that for several SKUs, more than 60–75% of reorder level consists of buffer stock rather than cycle stock.

This suggests that:

- Inventory structure is risk-driven rather than demand-driven.
- High variability directly inflates capital commitment. Class A items with moderate variability exert the strongest influence on working capital.
- Reducing demand uncertainty (via forecasting improvement or lead-time compression) would mathematically reduce safety stock requirements and improve capital efficiency.

4.4 Inventory Policy Recommendations

Based on integrated findings, the following policy framework is recommended:

4.4.1 Differentiated Replenishment System

- Continuous review (Q-system) for Class A volatile SKUs
- Periodic review (P-system) for Class B and C
- Reduced monitoring for stable low-risk SKUs

4.4.2 Dynamic Safety Stock Adjustment

- Increase monitoring frequency for Rising SKUs
- Reduce buffers gradually for Declining SKUs
- Recalculate CV quarterly

4.4.3 Lead-Time Reduction Strategy

- Since safety stock is proportional to variability and lead time:
- Reducing supplier lead time will directly reduce buffer requirement.

- Supplier negotiation and delivery scheduling should be explored.

4.4.4 Capital Efficiency Monitoring

Introduce KPI:

- Safety Stock / Average Demand ratio
- If ratio exceeds 2, conduct policy review.
- This institutionalises inventory discipline.

4.5 Strategic Conclusion

The integrated ABC, variability, trend, and safety stock analysis demonstrates that inventory optimisation at A.R. Furniture requires a risk-adjusted, differentiated control strategy.

Financial importance alone is insufficient for prioritisation. Demand uncertainty and trend direction significantly influence working capital exposure and service performance.

A structured, data-driven replenishment framework balancing service level and capital efficiency will enhance resilience and operational performance.